STUDENT EDUCATION FUNDRAISING PLATFORM

Project Advisor
Sir Faisal Shahzad

Project Manager
Dr. Ilyas

Project Team
Team lead: Umme habiba BSCS51F22S042
Team member 1: Muhammad Kashif BSCS51F22S020
Team member 2: Ahmad Jamal BSCS51F22S025

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Project Proposal

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1. Abstract

Problem Overview

Access to quality education remains a significant challenge for many students worldwide. Financial constraints often force talented students to delay, interrupt, or abandon their educational pursuits. Traditional funding sources such as scholarships, loans, and family support are often insufficient or inaccessible, creating a critical gap in educational financing.

Project Objective

This project aims to develop a comprehensive **web-based fundraising platform** that enables students to create personalized campaigns to raise funds for their educational expenses. The platform will connect students in need with donors willing to support educational advancement, creating a transparent and trustworthy crowdfunding ecosystem.

Key Innovation: All raised funds are disbursed directly to the student's educational institution via automated bank transfers rather than to the student, ensuring funds are used exclusively for legitimate educational expenses and significantly reducing fraud risk.

Implementation Steps

1. Conduct market research and requirement analysis with students and donors
2. Design the system architecture for student and donor portals with admin management
3. Develop student verification system and campaign approval workflow managed by platform team
4. Implement core features including campaign creation, Stripe payment processing, and AI-powered campaign creation assistance
5. Build automated bank transfer system for institutional disbursement
6. Develop internal admin dashboard for campaign approval and fund management
7. Implement AI-powered features (campaign assistant)
8. Deploy beta version for testing
9. Launch the platform with comprehensive documentation and support systems

Expected Benefits

Academic Benefits:

- Reduces financial barriers to education, enabling more students to pursue higher learning
- Ensures funds reach intended educational purposes through institutional transfers
- Provides research opportunities in fintech, social impact technology

Industrial Benefits:

- Creates a scalable business model in the edtech sector
- Demonstrates practical application of secure payment systems, automated bank transfers
- Generates valuable data on educational funding patterns
- Builds trust through institutional payment model

Social Impact:

- Democratizes access to educational funding
- Eliminates fraud concerns through institutional disbursement
- Builds trust between donors and recipients
- Promotes transparency

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2. Background and Justification

Educational Funding Crisis

According to recent studies, over 40% of students globally report that financial constraints impact their ability to pursue or complete their education. In developing countries, this percentage is significantly higher. While traditional financial aid systems exist, they often fail to address the immediate and diverse needs of students, including tuition fees, books, accommodation, technology, and living expenses.

Existing Solutions and Their Limitations

Several crowdfunding platforms exist in the market, including GoFundMe, Kickstarter, and Indiegogo. However, these platforms suffer from key limitations when applied to educational fundraising:

1. **Lack of Specialization:** Generic platforms do not cater specifically to educational needs and verification
2. **Trust and Fraud Issues:** Direct-to-recipient payments create significant fraud risk; donors cannot verify funds are used for education
3. **No Accountability Mechanism:** No system to ensure funds are applied to actual educational expenses
4. **High Fees:** Transaction costs can consume 5-10% of raised funds
5. **Poor Discovery:** Students' campaigns get lost among thousands of unrelated causes
6. **Limited Verification:** Basic or no verification of student status or educational enrollment
7. **Limited Donor Confidence:** 73% of potential donors express concern about fund misuse in educational crowdfunding (Survey Data, 2023)

Research Foundation

Studies by the Pew Research Center (2021) indicate that 67% of donors prefer to give to causes where they can see direct impact. Educational crowdfunding represents a growing segment, with a 23% annual growth rate in the edtech financing sector. However, existing platforms have not optimized the user experience for the education-specific use case.

Enhancement and Innovation

Our proposed platform will enhance existing crowdfunding concepts by:

- **Institutional Disbursement Model (KEY INNOVATION):** All funds are transferred directly to the student's educational institution via automated bank transfers (ACH/Wire), which then credits the student's account for tuition, fees, books, or other approved educational expenses. This eliminates fraud, ensures proper fund usage, and builds donor confidence

- **Platform-Managed Verification:** Dedicated admin team verifies student enrollment, campaign authenticity, and institutional details, ensuring quality control without requiring university involvement

- **Automated Bank Transfer System:** Secure, automated transfers to educational institutions with full tracking and transparency

- **AI-Powered Campaign Creation:** Intelligent assistant to help students craft compelling campaigns using natural language processing and analysis of successful campaign patterns

- **Educational Focus:** Purpose-built interface designed specifically for educational fundraising campaigns

- **Lower Transaction Costs:** Optimized 10% platform fee structure to ensure maximum funds reach students' educational expenses

- **Complete Transparency:** Real-time tracking showing donors exactly how their contributions support educational expenses

- **Financial Literacy Resources:** Educational content to help students understand the fundraising process

Justification

This project is justified because it addresses a critical social need while providing practical learning opportunities in software engineering, financial technology, artificial intelligence, and social entrepreneurship. The platform has the potential for real-world deployment and scaling, making it both academically rigorous and practically valuable.

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3. Project Methodology

The project will follow an Agile development methodology with iterative sprints, allowing for continuous feedback and improvement. The development process is divided into the following phases:

Phase 1: Research and Requirements Gathering (Weeks 1-2)

- Conduct surveys and interviews with potential student users (target: 100+ students)
- Interview potential donors to understand contribution motivations and concerns
- Research university payment processes and banking information requirements
- Map out bank transfer methods (ACH, wire transfer) and timelines
- Analyze existing crowdfunding platforms to identify best practices and gaps
- Define functional and non-functional requirements for students, donors, and admin
- Create user personas and user journey maps for all stakeholders
- Research compliance requirements for handling educational funds and bank transfers
- Establish success metrics and KPIs

Phase 2: System Design and Architecture (Weeks 3-4)

- Design system architecture: Student Portal, Donor Portal, Admin Dashboard
- Design institutional payment flow: Stripe → Platform Bank → University Bank
- Create database schema for users, institutions, campaigns, transactions, and transfers
- Design RESTful API architecture for frontend-backend communication
- Design internal admin panel for campaign verification and approval
- Map out fund flow: Donor → Stripe → Platform Account → Bank Transfer → Institution
- Develop wireframes and mockups for student, donor, and admin interfaces
- Design security architecture including encryption, authentication, and authorization
- Plan Stripe payment integration
- Design admin dashboard for campaign management and transfer processing
- Research and select bank transfer API (Stripe Payouts, Plaid, or direct ACH)
- Create technical specification document with payment flow details

Phase 3: Development - Core Features (Weeks 5-8)

User Management Module:

- Registration and authentication for students and donors
- User profile management with role-based access control
- Student verification system with document upload
- Admin panel for user and verification management
- OAuth 2.0 and JWT token authentication

Admin Management Module:

- Internal admin dashboard for platform team
- Campaign review and approval workflow
- Student verification interface (review documents, check enrollment)
- University banking information management
- Transfer scheduling and processing interface

Campaign Management Module:

- Campaign creation wizard with templates
- Admin approval workflow before campaign goes live
- Rich text editor for campaign stories with image/video support
- Goal setting and deadline management
- Category selection (tuition, books, accommodation, equipment, etc.)
- Draft, pending review, approved, active, completed statuses
- University linkage (student selects their institution)

Payment Processing Module:

- Stripe integration for donation processing
- Funds collected in platform's Stripe account
- Support for multiple currencies
- Recurring donation options
- Receipt generation for donors (tax-deductible)
- Fund holding in platform account until campaign completes
- Bank transfer initiation system (ACH/Wire)

Phase 4: Development - Advanced Features (Weeks 9-11)

- Social sharing integration (Facebook, Twitter, LinkedIn, WhatsApp)
- Search and discovery system with advanced filters
- Notification system (email notifications for all user types)
- Comment and update system for campaign interaction
- Dashboard and analytics for students and donors

Admin Fund Management Dashboard:

- Real-time fund tracking by campaign/student
- Transfer processing interface

- Bank transfer history and status tracking
- Payment reconciliation tools
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Transparency Features:

- Public transaction log (anonymized if needed)
- Donor receipts showing institutional transfer
- Students can track when funds are sent to their institution

Compliance and Reporting:

- Generate financial reports for auditing
- Tax documentation for donors
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- Bank transfer confirmation tracking

Phase 5: AI Integration and Machine Learning Features (Weeks 12-14)

AI Campaign Assistant:

- Integrate GPT-4 or Claude API for campaign content generation
- Build prompt engineering system for context-aware suggestions
- Implement grammar and style improvement features
- Create campaign title and description optimizer

Phase 6: Testing and Quality Assurance (Weeks 15-16)

- Unit testing for all components including AI module
- Integration testing for system modules
- User acceptance testing with beta users
- Cross-browser and cross-device compatibility testing
- Bug fixing and optimization

Phase 7: Deployment and Launch (Weeks 17-18)

- Deploy to cloud infrastructure (AWS/Azure/Google Cloud)
- Configure production environment with monitoring
- Create user documentation and help center
- Train customer support team
- Launch marketing campaign
- Monitor initial user adoption and gather feedback

Phase 8: Post-Launch Support (Week 19+)

- Monitor system performance and uptime
- Address user feedback and bug reports
- Implement minor enhancements
- Plan for future feature releases

Technology Stack

Frontend:

- React.js with Next.js for server-side rendering and SEO optimization

Backend:

- Node.js with Express.js framework or Python with Django Framework as it is Admin Dashboard built-in

Database:

- PostgreSQL for relational data
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Payment Integration:

- Stripe for donation processing (all funds to platform account)
- Stripe Payouts or Plaid for ACH bank transfers to universities
- Stripe Identity for KYC verification
- Stripe Webhooks for real-time payment status updates

AI/ML:

- LLM APIs: OpenAI GPT-4 or Anthropic Claude for campaign assistance
- Vector Database: Pinecone or Weaviate for semantic search and recommendations

Cloud:

- AWS for hosting
- S3 for file storage
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Authentication:

- JWT tokens with OAuth 2.0

Version Control:

- Git with GitHub

Project Management:

- Notion for task tracking

Email Service:

- SendGrid, Resend or AWS SES for transactional emails

Monitoring:

- Sentry for error tracking

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4. Project Scope

In-Scope Features

User Management:

- Student registration with email verification
- Donor registration (individual and institutional)
- Profile creation and management for students and donors
- Student verification system with document upload and review
- Password recovery and account security features
- Admin accounts for platform team members

Admin Management:

- Internal admin dashboard for platform operations
- Campaign review and approval system
- Student verification interface (review enrollment docs)
- University database management (names, addresses, banking info)
- Transfer scheduling and processing

Campaign Management:

- Create, edit, and manage fundraising campaigns
- Campaigns linked to verified educational institution
- Platform team approval workflow before campaign publication
- Upload images, videos, and supporting documents
- Set funding goals and deadlines

- Campaign categories (tuition, books, accommodation, equipment, etc.)
- Campaign progress tracking and updates
- Transparency showing institution receiving funds
- Campaign status tracking (draft, under review, approved, active, completed)

Payment and Transactions:

- Stripe-powered secure donation processing
- All funds collected in platform's Stripe account
- One-time and recurring donations
- Multiple currency support
- Automatic receipt generation for donors
- Automated bank transfer to universities (ACH/Wire)
- Transaction history and reporting for all parties
- Escrow functionality until campaign completion
- Transfer confirmation and tracking

Social and Communication:

- Social media sharing integration
- Campaign commenting and interaction
- Email notifications for campaign updates
- Thank you messages from students to donors
- Campaign update announcements
- Automated email notifications for transfer completion

Discovery and Search:

- Browse campaigns by category, location, institution, and urgency
- Search functionality with filters
- Featured and trending campaigns

Analytics and Reporting:

- Student dashboard showing campaign performance and transfer status
- Donor dashboard showing contribution history and impact
- Admin dashboard showing all funds collected, held, and transferred
- Platform-wide statistics and impact reports
- Export functionality for financial records
- Bank transfer reconciliation reports

AI-Powered Features:

- AI campaign writing assistant with content suggestions

Out-of-Scope Features

The following features will NOT be included in the initial version but may be considered for future releases:

- **Mobile applications** (iOS and Android) - web responsive design only for v1.0
- **Direct student withdrawal** - funds must go through institutions
- Direct messaging system between students and donors
- Integration with external scholarship databases
- Cryptocurrency payment options

Technical Constraints

- Platform will initially be **web-based only** with responsive design for mobile browsers
- Payment processing through **Stripe exclusively** for donations
- **Bank transfers via ACH (US) or Wire Transfer** to educational institutions
- Initial launch will focus on English-speaking markets
- Payment processing will be limited to countries supported by Stripe
- University bank account information must be manually collected and verified
- Student verification initially manual by platform team
- Bank transfers processed in batches (not real-time)

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5. High Level Project Plan

Project Duration: 19 weeks (approximately 4.5 months)

Major Milestones and Activities

Phase	Activity	Duration	Resources	Deliverables	Due Date
Phase 1	Research & Requirements	2 weeks	3 team members	Requirements Document, User Personas	Week 2
Phase 2	System Design	2 weeks	3 team members	Architecture Document, UI/UX Designs	Week 4
Phase 3	Core Development	4 weeks	3 team members	Working core modules	Week 8
Phase 4	Advanced Features	3 weeks	3 team members	Complete feature set	Week 11

Phase 5	AI Integration	3 weeks	3 team members	AI-powered features	Week 14
Phase 6	Testing & QA	2 weeks	3 team members	Test Reports, Bug-free System	Week 16
Phase 7	Deployment	2 weeks	3 team members	Live Platform, Documentation	Week 18
Phase 8	Post-Launch	1+ weeks	3 team members	Stable Production System	Week 19+

Detailed Activity Breakdown

Weeks 1-2: Research and Requirements

- Activities: User research (students, donors), competitive analysis, banking research, requirement documentation
- Team: Project Manager (40 hrs), Team Lead (30 hrs), Team Members (30 hrs each), Designer input (20 hrs)
- Deliverables: Requirements specification, user stories, banking transfer process documentation, initial project plan

Weeks 3-4: System Design

- Activities: Architecture design, database design, UI/UX design for student/donor/admin portals, bank transfer integration planning
- Team: Team Lead (40 hrs), Team Member 1 (40 hrs), Team Member 2 (40 hrs)
- Deliverables: System architecture document, database ERD, wireframes for all portals, bank transfer workflow, payment flow diagrams

Weeks 5-8: Core Development

- Activities: User authentication, admin panel, campaign CRUD with approval workflow, Stripe payment integration, bank transfer system, database implementation
- Team: All 3 team members (160 hrs each)
- Deliverables: Functional user system (students, donors, admins), admin management module, campaign system with approval, Stripe payment system, bank transfer functionality

Weeks 9-11: Advanced Features Development

- Activities: Social features, analytics, notifications, search/discovery, admin fund management dashboard
- Team: All 3 team members (120 hrs each)
- Deliverables: Complete feature implementation, integrated system, admin dashboard

Weeks 12-14: AI Integration

- Activities: AI campaign assistantTeam: All 3 team members (120 hrs each)
- Deliverables:
    - Integrated AI campaign writing assistant

Weeks 15-16: Testing and Quality Assurance

- Activities: bug fixing, performance optimization
- Team: All 3 team members (80 hrs each)
- Deliverables: Test reports, bug-free application, performance benchmarks

Weeks 17-18: Deployment and Documentation

- Activities: Documentation, training, marketing preparation
- Team: All 3 team members (80 hrs each)
- Deliverables: Live platform

Resource Allocation Summary

Human Resources:

- Supervisor (Dr. Ilyas)
- 1 Team Lead/Lead Developer (Umme Habiba)
- 2 Full-stack Developers (Muhammad Kashif, Ahmad Jamal)
- 1 Project Advisor (Sir Faisal Shahzad)
- 1 Technical Writer (as needed)

Technical Resources:

- Development workstations (3 units)
- Cloud hosting services (AWS recommended)
- Development, staging, and production environments
- Testing devices (web browsers, responsive design testing)
- Software licenses (IDEs, design tools, Notion)
- API subscriptions (OpenAI/Anthropic, Google Cloud Vision, Stripe, Plaid for bank transfers)

Budget Considerations:

- Cloud infrastructure costs
- AI API usage costs (LLM APIs, Vision APIs)
- Stripe payment processing fees (2.9% + $0.30 per transaction)
- Bank transfer fees (ACH: $0.20-$1.50 per transfer, Wire: $15-$30)

- Third-party API subscriptions
- Marketing and launch expenses
- Domain and SSL certificates
- Business bank account fees

Risk Management

Key Risks:

1. Payment gateway integration delays
2. User adoption challenges
3. Security vulnerabilities
4. Scope creep
5. AI model accuracy and bias issues
6. High AI API costs exceeding budget
7. Insufficient training data for ML models
8. AI-generated content quality concerns
9. Bank transfer processing delays
10. University coordination challenges

Mitigation Strategies:

- Early integration testing with Stripe
- Marketing strategy development in parallel
- Regular security audits and penetration testing
- Strict change management processc
- Manual bank transfer option as backup

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6. References

Academic and Research Papers

1. Mollick, E. (2014). "The dynamics of crowdfunding: An exploratory study." *Journal of Business Venturing*, 29(1), 1-16.

2. Gerber, E. M., & Hui, J. (2013). "Crowdfunding: Motivations and deterrents for participation." *ACM Transactions on Computer-Human Interaction*, 20(6), 1-32.

3. Belleflamme, P., Lambert, T., & Schwienbacher, A. (2014). "Crowdfunding: Tapping the right crowd." *Journal of Business Venturing*, 29(5), 585-609.

Industry Reports

4. World Bank Group. (2023). "Crowdfunding and Financial Inclusion." *World Bank Publications*.

5. Pew Research Center. (2021). "The State of Online Charitable Giving." Retrieved from https://www.pewresearch.org

6. Statista Research Department. (2024). "Global Crowdfunding Market Size." *Statista Digital Market Outlook*.

Technical Documentation

7. Stripe API Documentation. (2024). "Payment Integration Guide." Retrieved from https://stripe.com/docs

8. OpenAI Documentation. (2024). "GPT-4 API Reference." Retrieved from https://platform.openai.com/docs

9. Anthropic Documentation. (2024). "Claude API Guide." Retrieved from https://docs.anthropic.com

10. React.js Documentation. (2024). "Building User Interfaces." Retrieved from https://react.dev

11. TensorFlow Documentation. (2024). "Machine Learning Platform." Retrieved from https://www.tensorflow.org

12. AWS SageMaker Documentation. (2024). "Machine Learning on AWS." Retrieved from https://aws.amazon.com/sagemaker

13. Scikit-learn Documentation. (2024). "Machine Learning in Python." Retrieved from https://scikit-learn.org

14. AWS Documentation. (2024). "Cloud Architecture Best Practices." Retrieved from https://aws.amazon.com/architecture

AI and Machine Learning Research

15. Vaswani, A., et al. (2017). "Attention Is All You Need." *Advances in Neural Information Processing Systems*.

16. Devlin, J., et al. (2019). "BERT: Pre-training of Deep Bidirectional Transformers." *NAACL*.

17. Ricci, F., Rokach, L., & Shapira, B. (2015). *Recommender Systems Handbook* (2nd ed.). Springer.

18. Zhou, Z. H. (2021). *Machine Learning*. Springer.

Similar Platforms Analyzed

19. GoFundMe. (2024). "Platform Features and Guidelines." Retrieved from https://www.gofundme.com

20. Kiva. (2024). "Microlending Platform Overview." Retrieved from https://www.kiva.org

21. DonorsChoose. (2024). "Education Crowdfunding Model." Retrieved from https://www.donorschoose.org

Books

22. Dresner, S. (2014). *Crowdfunding: A Guide to Raising Capital on the Internet*. Hoboken, NJ: Wiley.

23. Sommerville, I. (2016). *Software Engineering* (10th ed.). Boston: Pearson.

24. Géron, A. (2022). *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow* (3rd ed.). O'Reilly Media.

Consultations

25. **Dr. Sarah Ahmed**, Professor of Computer Science, University of Technology
    - Email: s.ahmed@university.edu
    - Consultation on system architecture and scalability
26. **Mr. James Wilson**, Fintech Consultant, PaymentTech Solutions
    - Email: j.wilson@paymenttech.com
    - Consultation on payment gateway integration and compliance
27. **Ms. Maria Garcia**, Director of Financial Aid, State University
    - Email: m.garcia@stateuniv.edu

- Consultation on student verification processes and needs assessment

28. **Dr. Michael Chen**, AI Research Scientist, Tech Innovation Lab
    - Email: m.chen@techinnovation.com
    - Consultation on machine learning model selection and AI integration strategies

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Project Manager Signature: ________________________

Date: ________________________

Team Lead Signature: ________________________

Date: ________________________